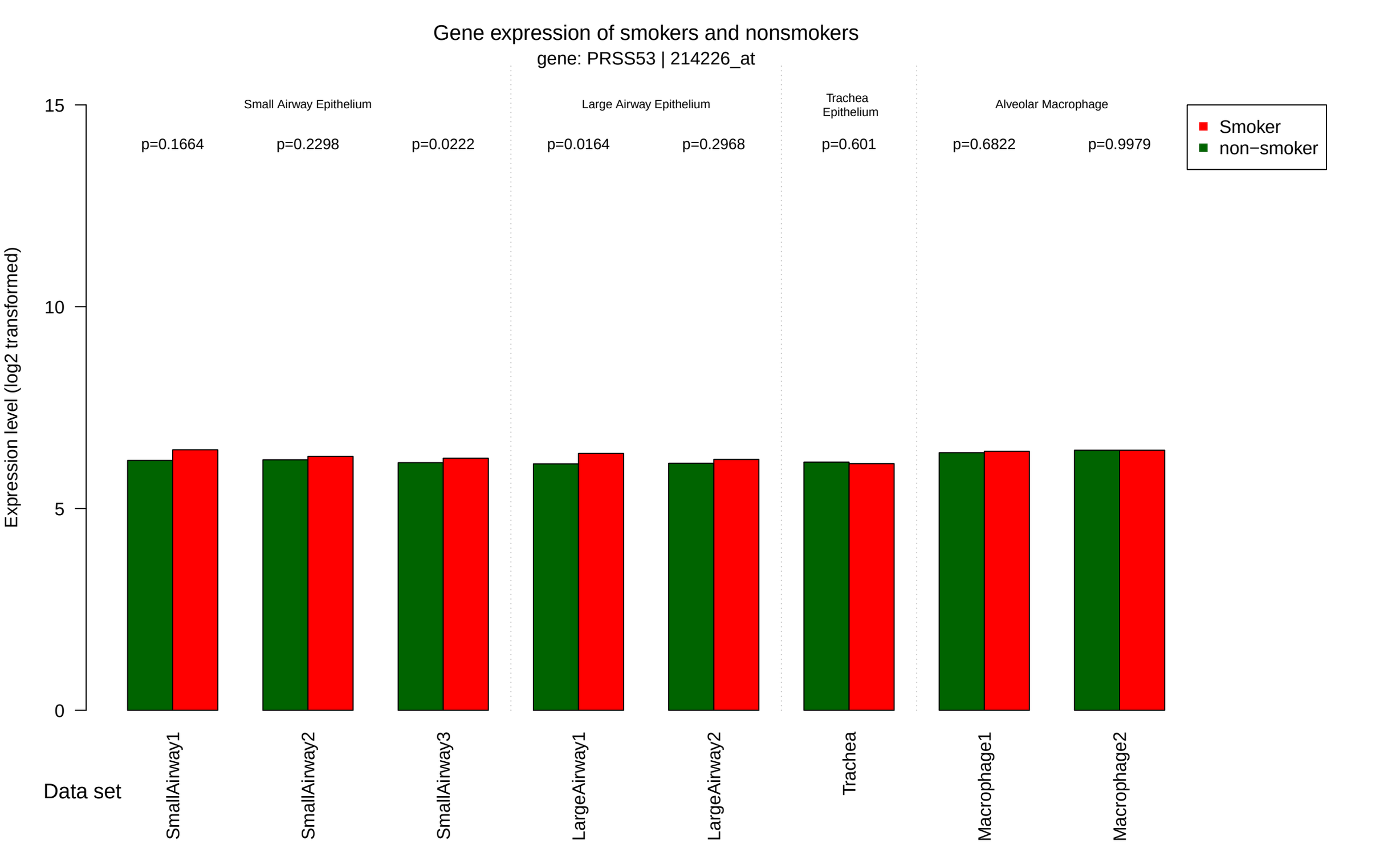
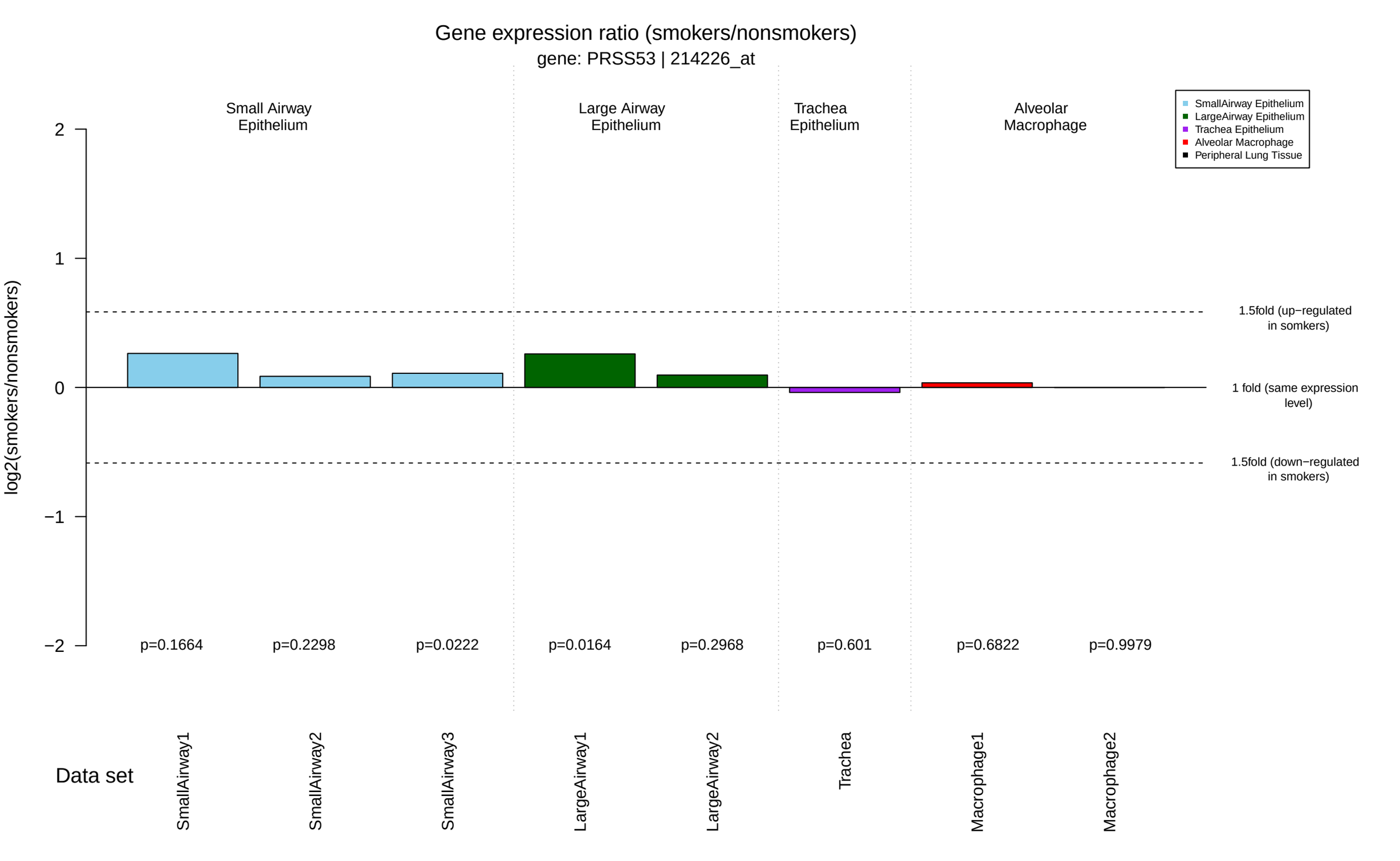
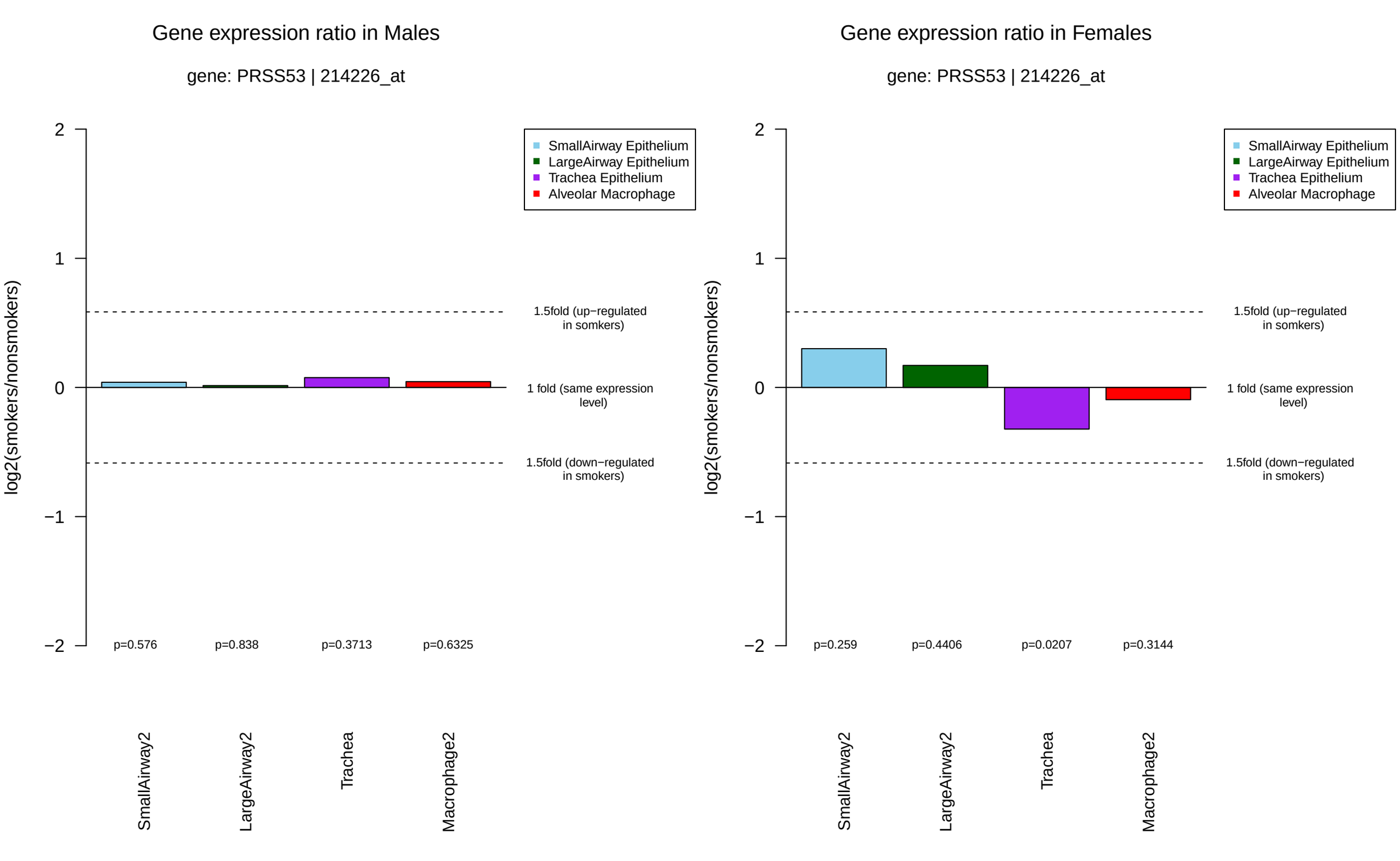
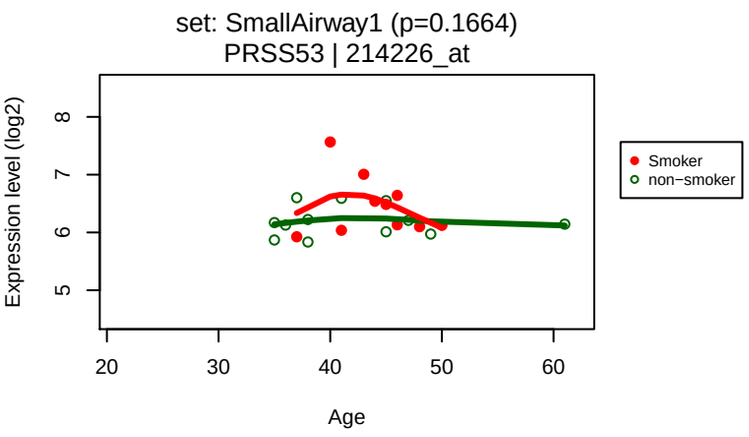
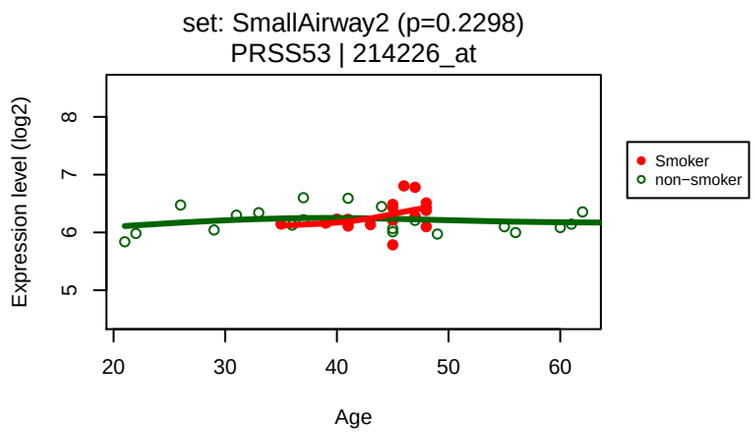
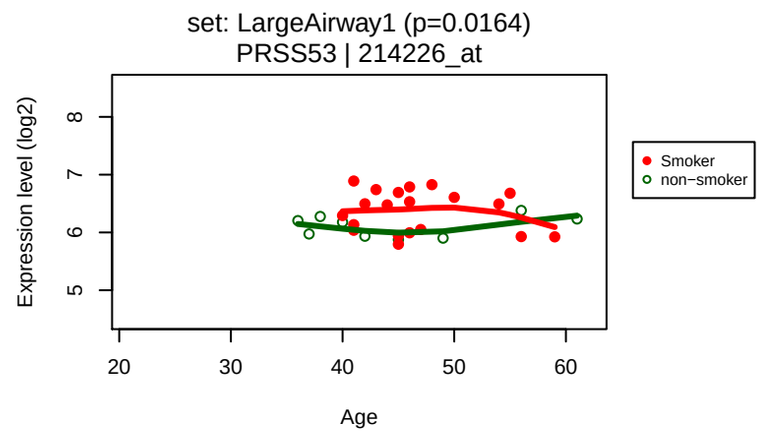

Gene expression of smokers and nonsmokers
gene: PRSS53 | 214226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1664
p=0.2298
p=0.0222
p=0.0164
p=0.2968
p=0.601
p=0.6822
p=0.9979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

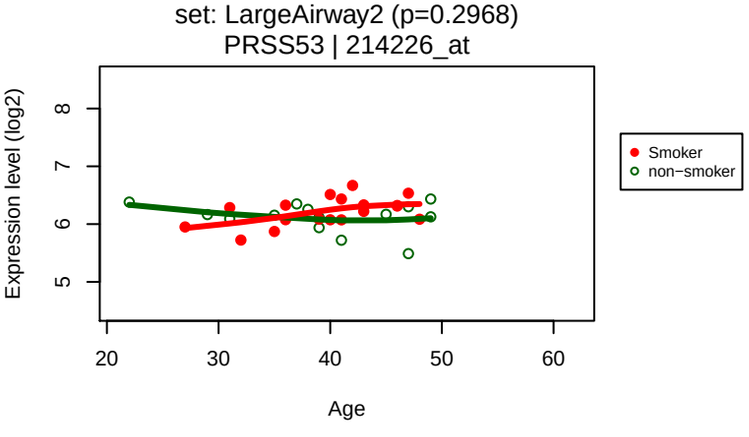

Gene expression ratio (smokers/nonsmokers)
gene: PRSS53 | 214226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1664
p=0.2298
p=0.0222
p=0.0164
p=0.2968
p=0.601
p=0.6822
p=0.9979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

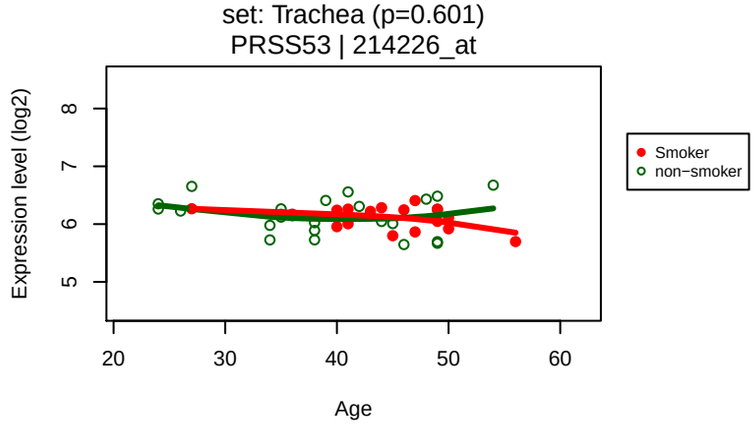

Gene expression ratio in Males
gene: PRSS53 | 214226_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.576
p=0.838
p=0.3713
p=0.6325
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS53 | 214226_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.259
p=0.4406
p=0.0207
p=0.3144
SmallAirway2
LargeAirway2
Trachea
Macrophage2

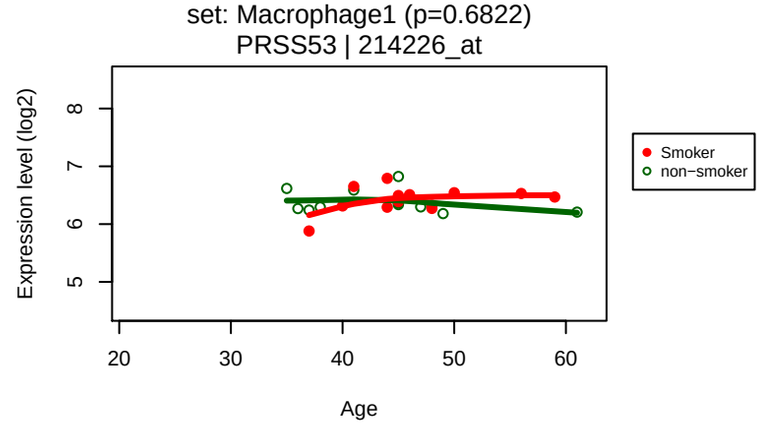

set: SmallAirway1 (p=0.1664)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker

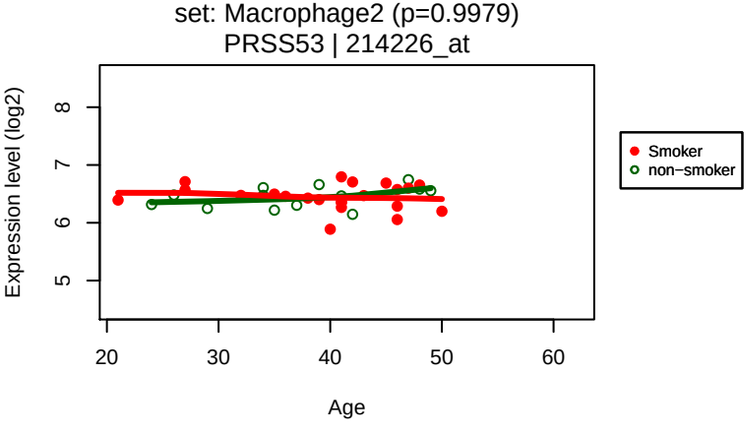

set: SmallAirway2 (p=0.2298)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker

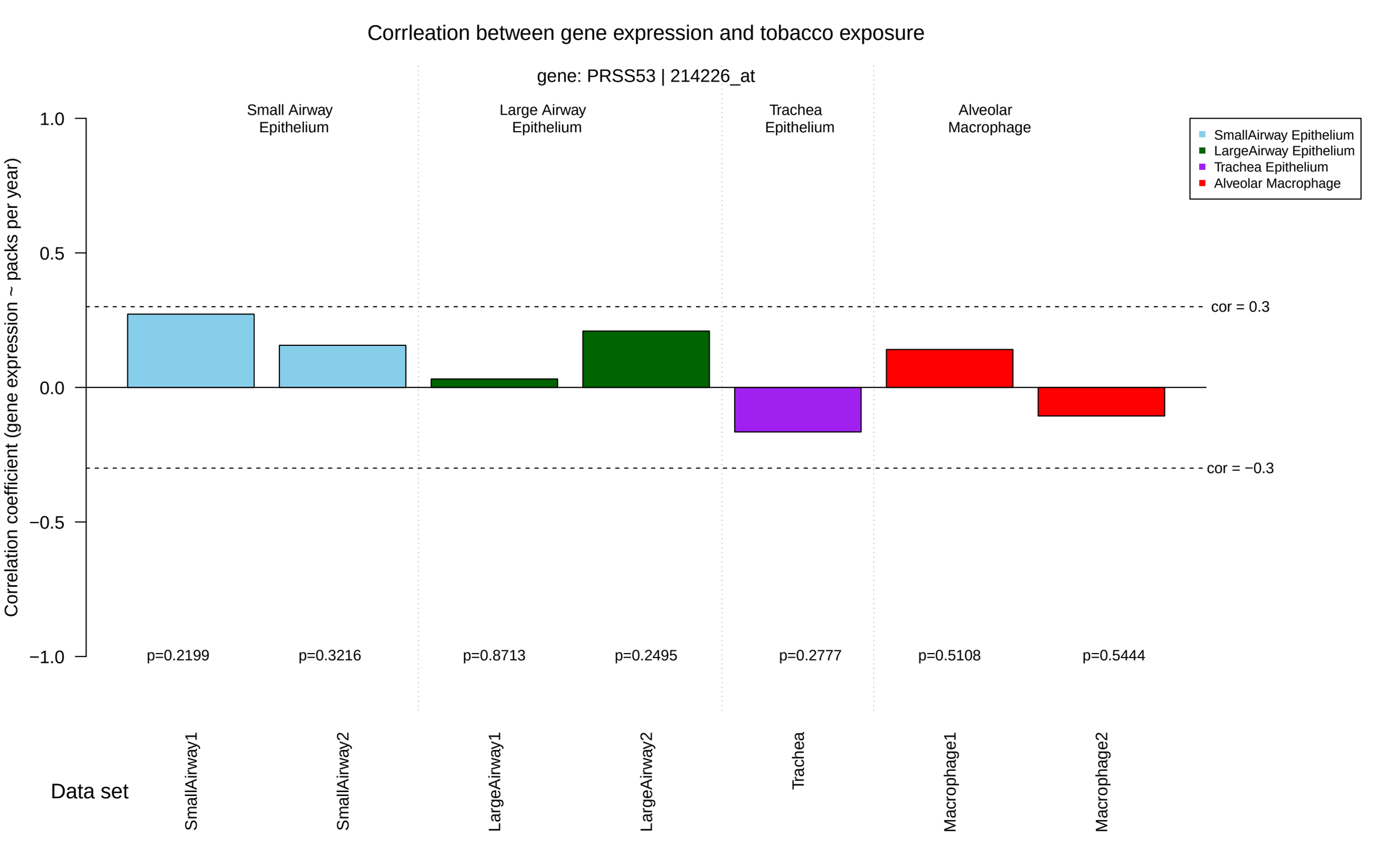

set: LargeAirway1 (p=0.0164)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2968)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.601)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6822)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9979)
PRSS53 | 214226_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS53 | 214226_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2199
p=0.3216
p=0.8713
p=0.2495
p=0.2777
p=0.5108
p=0.5444
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2